In [1]:
```
import pandas as pd
product_info=pd.Series({'keyboard':560,'Mouse':700,'Monitor':500,'tablet':800,'prin
print("product with available quantity:")
print(product_info)
print("-------------------------------------------")
print("quantity of mouse:",product_info['Mouse'])
```

```
product with available quantity:
keyboard    560
Mouse       700
Monitor     500
tablet      800
printer     500
dtype: int64
-------------------------------------------
quantity of mouse: 700
```

In [2]:
```
#---------------------------
# creating a series
data=pd.Series([78,90,45,90,79],index=['Hindi','Science','English','Maths','Compute
#to sort the data as per values
print(data.sort_values())
# to sort data as per index
print("------------------------------")
print("Sorted data as per index:")
print(data.sort_index())
```

```
English     45
Hindi       78
Computer    79
Science     90
Maths       90
dtype: int64
------------------------------
Sorted data as per index:
Computer    79
English     45
Hindi       78
Maths       90
Science     90
dtype: int64
```

In [3]:
```
data=pd.Series([50,60,80,None,90,50,None])
print(data.isnull())
```

```
0    False
1    False
2    False
3     True
4    False
5    False
6     True
dtype: bool
```

In [4]:
```
import numpy as np
data1=pd.Series([50,60,None,90,30,np.nan,90,60,None])
#to fill empty values with 0
print(data1.fillna(0))
```

```
0    50.0
1    60.0
2     0.0
3    90.0
4    30.0
5     0.0
6    90.0
7    60.0
8     0.0
dtype: float64
```

In [5]:
```
data2=pd.Series([50,60,None,60,90,None,20,None])
#missing values treatment by filling data as average value
print(data2.fillna(data2.mean()))
```

```
0    50.0
1    60.0
2    56.0
3    60.0
4    90.0
5    56.0
6    20.0
7    56.0
dtype: float64
```

In [6]:
```
data3=pd.Series([60,90,None,50,34,None,None,78])
#dropping missing value
print(data3.dropna())
```

```
0    60.0
1    90.0
3    50.0
4    34.0
7    78.0
dtype: float64
```

In [7]:
```
#creating a series with Dictionary
product_info=pd.Series({'Keyboard':560,'Mouse':700,'Monitor':500,'Tablet':800,'Prin
#to display series
print("Product with Available quantity:")
print(product_info)
```

```
Product with Available quantity:
Keyboard    560
Mouse       700
Monitor     500
Tablet      800
Printer     500
dtype: int64
```

In [8]:
```
#to increment quantity of each product by 200
product_info + 200
```

```
print("Product with Available quantity:")
print(product_info)
```

```
Product with Available quantity:
Keyboard    560
Mouse       700
Monitor     500
Tablet      800
Printer     500
dtype: int64
```

In [9]:

```
#creating a series
marks=pd.Series([1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,75,65,45,67,54,33,44,
print("Series is :")
print (marks)
print("-------------------------------------------------------")
#to display first 5 values
print("First 5 values:")
print(marks.head(5))
```

```
Series is :
0      1.0
1      2.0
2      3.0
3      4.0
4      5.0
5      6.0
6      7.0
7      8.0
8      9.0
9     10.0
10    11.0
11    12.0
12    13.0
13    14.0
14    15.0
15    16.0
16    17.0
17    18.0
18    75.0
19    65.0
20    45.0
21    67.0
22    54.0
23    33.0
24    44.0
25    55.0
26     5.0
27    67.0
28     4.0
29     3.0
30    44.0
31    55.0
32    33.0
33    33.0
34    19.0
35    20.0
36     NaN
37    67.0
38     NaN
39    67.0
40    67.0
41     NaN
42    66.0
dtype: float64
-----------------------------------------------------
First 5 values:
0    1.0
1    2.0
2    3.0
3    4.0
4    5.0
dtype: float64
```

In [10]:
```
import pandas as pd
```

```python
# creating a series with dictionary
product_info = pd.Series({
    'Keyboard': 560,
    'Mouse': 700,
    'Monitor': 500,
    'Tablet': 800,
    'Printer': 500
})

# to display
print("Product with available quantity:")
print(product_info)

print("----------------------------")
print("Quantity of mouse:", product_info['Mouse'])
print("============================")
print("Quantity of keyboard:", product_info['Keyboard'])
```

```
Product with available quantity:
Keyboard    560
Mouse       700
Monitor     500
Tablet      800
Printer     500
dtype: int64
----------------------------
Quantity of mouse: 700
============================
Quantity of keyboard: 560
```

In [11]:

```python
#operations on Series
#to increment the quantity of each product by 200
product_info=product_info+200
print(product_info)
```

```
Keyboard     760
Mouse        900
Monitor      700
Tablet      1000
Printer      700
dtype: int64
```

In [12]:

```python
# to import pandas library
import pandas as pd

# to create a series
marks = pd.Series([89, 90, 70, 40, 65])

# to display the series
print(marks)

print("=============================")

# new data
data = [80, 90, 70, 45, 65]

# Indexing
```

```
subname = ['Science', 'Maths', 'Hindi', 'English', 'Computer']

marks = pd.Series(data, index=subname)
```

```
0    89
1    90
2    70
3    40
4    65
dtype: int64
==============================
```

In [13]:
```
# displaying marks
print(marks)
```

```
Science     80
Maths       90
Hindi       70
English     45
Computer    65
dtype: int64
```

In [14]:
```
#display the product having quantitygreater than 800
print("Products with quantity greater than 800")
print(product_info[product_info >800])
```

```
Products with quantity greater than 800
Mouse      900
Tablet    1000
dtype: int64
```

In [15]:
```
#creating a series
marks=pd.Series([
78,85,90,66,72,88,91,55,60,77,
82,69,74,88,92,45,58,63,79,84,
None,73,68,81,76,89,94,52,61,70,
87,93,64,59,80,75,None,67,83,71,
])
#-----------------------------------------
# to display first 5 values
print("First 5 values:")
print(marks.head(5))
print("------------------------------------")
# to display last 7 values
print("Last 7 values:")
print(marks.tail(7))
#to count the frequency of each value
print("Frequency of each value")
print(marks.value_counts())
```

```
First 5 values:
0    78.0
1    85.0
2    90.0
3    66.0
4    72.0
dtype: float64
------------------------------------
Last 7 values:
33    59.0
34    80.0
35    75.0
36     NaN
37    67.0
38    83.0
39    71.0
dtype: float64
Frequency of each value
88.0    2
78.0    1
85.0    1
90.0    1
66.0    1
72.0    1
91.0    1
55.0    1
60.0    1
77.0    1
82.0    1
69.0    1
74.0    1
92.0    1
45.0    1
58.0    1
63.0    1
79.0    1
84.0    1
73.0    1
68.0    1
81.0    1
76.0    1
89.0    1
94.0    1
52.0    1
61.0    1
70.0    1
87.0    1
93.0    1
64.0    1
59.0    1
80.0    1
75.0    1
67.0    1
83.0    1
71.0    1
Name: count, dtype: int64
```

In [16]:
```
import pandas as pd

# Step 1: 50 numeric values (with duplicates)
data = [
    1,2,3,4,5,2,3,4,5,1,
    6,7,8,9,10,2,3,4,5,6,
    7,8,9,10,1,2,3,4,5,6,
    7,8,9,10,1,2,3,4,5,6,
    7,8,9,10,1,2,3,4,5,6
]

df = pd.DataFrame(data, columns=["Numbers"])

# Step 2: Frequency check
print("Frequency Count:\n")
print(df["Numbers"].value_counts())

# Step 3: head() - first 5 values
print("\nFirst 5 values:\n")
print(df.head())

# Step 4: tail() - last 5 values
print("\nLast 5 values:\n")
print(df.tail())
```

```
Frequency Count:

Numbers
2     6
3     6
4     6
5     6
1     5
6     5
7     4
8     4
9     4
10    4
Name: count, dtype: int64

First 5 values:

   Numbers
0        1
1        2
2        3
3        4
4        5

Last 5 values:

    Numbers
45        2
46        3
47        4
48        5
49        6
```

In [17]:
```
#to display unique values
print("Unique values:")
print(marks.unique())
#total count of unique values
print("-----------------------------------------------")
print("Total Unique values:",marks.nunique())
```

```
Unique values:
[78. 85. 90. 66. 72. 88. 91. 55. 60. 77. 82. 69. 74. 92. 45. 58. 63. 79.
 84. nan 73. 68. 81. 76. 89. 94. 52. 61. 70. 87. 93. 64. 59. 80. 75. 67.
 83. 71.]
-----------------------------------------------
Total Unique values: 37
```

In [18]:
```
data3=pd.Series([60,90,None,50,34,None,None,78])
#dropping missing value
print(data3.dropna())
```

```
0    60.0
1    90.0
3    50.0
4    34.0
7    78.0
dtype: float64
```

In [ ]:

```

```